UU-P-221B
December 26, 1972
SUPERSEDING
Fed. Spec. UU-P-221A
October 27, 1964

FEDERAL SPECIFICATION

PAPER, DIRECT-POSITIVE SENSITIZED (DIAZOTYPE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers diazo sensitized paper for producing direct-positive prints from translucent or transparent originals by contact printing and by processing in ammonia vapor, moist or other systems designed for development of these products.

1.2 Classification. Diazotype papers covered by this specification shall be classified using the designators specified in table I, and meeting the requirements put forth in table II. The paper products nominal value of the appropriate characteristic will fall within one of the designator groups. The paper product supplied by the manufacturer shall not vary more than ±10 percent of the manufacturer's nominal value. While this designator system includes every conceivable combination of sensitized diazo-positive paper product, many of these products are not practically possible in the diazo process at this time.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-I-00548 - Ink, Black, for Offset Duplicating Process.
PPP-T-60 - Tape; Pressure-sensitive Adhesive, Waterproof, for Packaging.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 131 - Diazotype Paper Sheet Sizes.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-P-3447 - Paper, Map, Packaging of.

Military Standards:

MIL-STD-2 - Drawing, Sizes and Formats.
MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Circular:

No. 547 - Preparation and Colorimetric Properties of Magnesium Oxide Reflectance Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington DC 20402.)

American National Standards Institute (ANSI), Inc. Standards:

No. PH2.17 - Diffuse Reflection Density.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

Technical Association of the Pulp and Paper Industry (TAPPI):

Paper, Methods of Testing

TAPPI NO.		TAPPI METHOD
T402-m-49	-	Conditioning of Paper and Paperboard for testing.
T410-os-68	-	Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T511-su-69	-	Folding Endurance of Paper (MIT Tester).
T414-ts-65	-	Internal Tearing Resistance of Paper.
T401-m-60	-	Fiber Analysis of Paper and Paperboard.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York, New York 10017.)

National Motor Freight Traffic Association, Inc., Agent:

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, IL, 60606.)

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3. REQUIREMENTS

3.1 Materials. The diazotype paper procured in accordance with this specification shall meet all the requirements specified herein. Any material which exhibits coating or physical defects not meeting the specification prior to the expiration date shall be replaced by the manufacturer at no additional cost to the government.

3.2 Manufacturing defects. When tested as specified, the expiration date of the diazotype paper shall not be less than 90 days from the date of shipment.

3.3 Expiration date. Unless otherwise specified, the expiration date of the diazotype paper shall not be less than 90 days from the date of shipment.

3.3.1 Preproduction sample. When specified (see 6.1.1), preproduction samples, as specified, shall be submitted for examination, test, and inspection in accordance with section 4. In all respects the preproduction samples shall represent the identical quality of materials workmanship and design which will be used in the manufacture of the diazotype papers.

3.4 Classification designators (see table I).

3.4.1 Type. Type 1 sensitized paper shall consist of a paper base coated with a light sensitive layer containing one or more diazo compounds and one or more couplers which shall be capable of the formation of a dye image when developed in a suitable ammoniacal atmosphere. Type 2 paper shall consist of a paper base coated with a light sensitive layer containing one or more diazo compounds which shall be capable of the formation of a dye image when developed with a standard alkaline developer solution. Type 3 paper shall consist of a paper base coated with a light sensitive layer containing one or more diazo compounds which shall be capable of the formation of a dye image when developed with a standard neutral developed solution. Type 4 paper shall consist of a paper base coated with a light sensitive diazo layer which shall be capable of the formation of a dye image when developed in the system certified by the manufacturer for use with type 4 material.

3.4.2 Style. Style 1 paper shall consist of a white paper base essentially opaque to actinic light. Style 2 paper shall consist of an actinically translucent or transparent paper base. Style 3 paper shall consist of an actinically translucent or tranparent paper base and coated with a light sensitive layer capable of forming an erasable dye,image as specified in 3.5.9.

3.4.3 Image color. The image color shall conform to the designator in table I; blue, black, red (maroon), or sepia (brownish color with an image of high actinic opacity).

3.4.4 Class. The paper stock for Classes 1, 2, 3, 4, and 5 papers shall be composed of chemically bleached wood fibers, cotton fibers (rag), or a combination of chemically bleached wood fibers and cotton fibers and meet the physical characteristics given in table II. Classes 6 and 7 paper stock shall be 100 percent rag and meet the physical characteristics given in table III. Classes 8 and 9 stock shall be composed of chemically bleached wood fibers, cotton fibers (rag), cotton linters or a combination thereof and meet the physical characteristics given in table II. All paper stock shall be free from unbleached fibers and ground wood pulp.

3.4.5 Sub class. Sub class 1 shall consist of a paper base coated one one side with a light sensitive layer capable of the formation of a dye image as specified. Sub class 2 shall consist of a paper base coated on both sides with light sensitive layers capable of the formation of a dye image as specified.

3.4.6 ARMM speed. The nominal ARMM speed designator of the paper shall conform to the values in table I when tested in accordance with 4.13. Speed variations from the manufacturer's nominal shall be within ± 10 percent.

3.4.7 Maximum density. The maximum density designator of papers specifed herein shall be according to the values in table I when tested, as specified in 4.14.1. Maximum density variations from the manufacturer's nominal shall be within ± 10 percent.

3.4.8 Minimum density. The minimum density designator of papers specified herein shall be according to values in table 1 when tested, as specified in 4.14.2. Minimum density variations from the manufacturer's nominal shall be within ± 10 percent.

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TABLE I

DIAZOTYPE PAPER CLASSIFICATION SYSTEM

Type *1	Style *2	Image Color *3	Class *1	Sub Class *2	ARMM Speed *1	*Density Maximum *3	Density Minimum *2
1. Dry Developing	1. Opaque	1. Blue	1. Light weight Opaque	1. Sensitized One side	1. less than 25	1. less than 0.90	1. less than 0.10
2. Moist Alkaline Developing	2. Reproducible	2. Black	2. Standard Weight Opaque	2. Sensitized Both sides	2. 25-40	2. 0.90-1.10	2. 0.10-0.19
3. Moist Neutral Developing	3. Erasable	3. Red	3. Medium weight Opaque		3. 41-50	3. 1.11-1.30	3. 0.20-0.30
4. Moist Pressure Developing		4. Sepia	4. Heavy weight Opaque		4. 51-70	4. 1.31-1.50	
			5. Card weight Opaque		5. 71-90	5. over 1.50	
			6. Light weight translucent (100 percent rag)		6. 91 - 115		
			7. Light weight transparent (100 percent rag)		7. 116-135		
			8. Light weight translucent		8. 136-170		
			9. Light Weight transparent		9. 171-210		
					10. Over 210		

* SAMPLE DESIGNATOR 1-2-3-1-2-1-3-2

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3.5 Product Characteristics (see table II).

3.5.1 Sizes.

3.5.1.1 Roll sizes. The rolls of sensitized paper shall be furnished in widths and lengths in accordance with MIL-STD-2 or as specified in the contract or purchase order. Unless otherwise specified, tolerance on roll widths shall be as stated in table II. The length of the paper shall be as stated in table II.

3.5.1.2 Splices in rolls. All splices shall be neatly made with not more than one splice in 50 yards of paper. At least one yard of sensitized paper shall be added to the roll for each splice. For nonroll feed paper, all splices shall be flagged at the splice with a colored marker projecting beyond the edge of the roll. For roll feed paper containing a splice, identification shall be printed on the outer paper wrap and not flagged. A single shipment shall not contain more than 10 percent of such spliced rolls.

3.5.1.3 Sheet sizes. Unless otherwise specified, sheets of sensitized paper shall be furnished in standard sizes specified in Fed. Std. No. 131. The sheets shall be cut square on four sides, and shall be cut to the sizes specified. The dimensions shall have tolerances as stated in table II. The sizes of the sheets and tolerances shall be measured immediately after cutting.

3.5.2 Physical characteristics. The physical characteristics of basis weight, fold endurance, and tear resistance shall be made in accordance with Technical Association Pulp and Paper Industry (TAPPI Methods) and meet the requirements as stated in table II. These tests shall be performed on the sensitized paper.

3.5.2.1 Temporary curl. Temporary curl of all papers shall not cause unsatisfactory stacking in front or rear of whiteprint machines when tested as specified in 4.10.

3.5.2.2 Permanent curl. All papers at the time of receipt of shipment shall have a permanent curl value not exceeding that given in table II when determined in accordance with 4.11.

3.5.2.3 Package curl. The paper shall be such that as it comes from the package it will be virtually flat and entirely suitable for rapid feeding into a whiteprint processor, and for handling by offset printing equipment.

3.5.3 Accelerated aging characteristics.

3.5.3.1 Change of maximum density of light aged samples. When tested, as specified in 4.14.3, the change of maximum density shall conform to table II.

3.5.3.2 Change of minimum density of light aged samples. When tested, as specified in 4.14.4, the change of minimum density shall conform to table II.

3.5.3.3 Change of maximum density of heat aged samples. When tested, as specified in 4.14.5.1, the change of maximum density shall conform to table II.

3.5.3.4 Change of minimum density of heat aged samples. When tested, as specified in 4.14.5.2, the change of minimum density shall conform to table II.

3.5.4 Actinic transmission. The actinic transmission shall conform to the value in table II when tested, as specified in 4.16.

3.5.5 Actinic opacity. The actinic opacity shall conform to the value in table II when tested, as specified in 4.15

3.5.6 Color. Unless otherwise specified, all paper stock specified herein shall be white.

3.5.7 Coating. The sensitized coating shall be uniform, essentially free from dirt, coating streaks, pinholes, abrasion marks, spots, mottling and areas of markedly higher or lower sensitivity which will affect print quality when tested, as specified in 4.12. The coating shall be sufficiently stable so that it will not come off on rollers in whiteprint machines or pile up on the blanket of offset printing presses, nor interfere with the imprint made by standard offset inks meeting TT-I-00548 (GPO), Ink, Black, For Offset Duplicating Process.

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3.5.8.1 Developing rate.

3.5.8.1 Developing rate for type 1 papers. The developing rate for type 1 papers shall be such that adequate development of the coating is attained in one pass through the developing section of the apparatus, as specified in 4.5.1.

3.5.8.2 Developing rate for type 2 papers. The developing rate for type 2 papers shall be such that adequate development of the coating is attained in one pass through the developing section of the apparatus, as specified in 4.5.2. Paper of this type shall be developed in a solution certified by the manufacturer for use with type 2 material.

3.5.8.3 Developing rate for type 3 papers. The developing rate for type 3 papers shall be such that adequate development of the coating is attained in one pass through the developing section of the apparatus, as specified in 4.5.3. Paper of this type shall be developed in a solution certified by the manufacturer for use with type 3 material.

TABLE II

PRODUCT CHARACTERISTICS

1. Format.

A. Rolls
B. Sheets

2. Sizes and Tolerances.

A. Rolls

Width	Length	Tolerance
In accordance with MIL-STD-2	In accordance with MIL-STD-2	Width: +3/8 inch -1/16 inch Length: Not less than nominal length

B. Sheets

Width	Length	Tolerance
In accordance with Fed. Std. No. 131.	In accordance with Fed. Std. No. 131	For sizes 8 X 10-1/2 inches through 15 X 18 inches inclusive: ±1/16 inch For sizes larger than 15 X 18 inches: ± 3/32 inch.

All sizes measured in atmospheric conditions to be in accordance with TAPPI Methods.

3. Physical characteristics.

A. Basis weight (17 X 22, 500 sheets) ± 7-1/2 percent.

Class	Pounds
1	17.5
2	20.0
3	24.0
4	32.0
5	40 to 60
6	14 to 20
7	15 to 20
8	15 to 20
9	15 to 20

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TABLE II (Continued)

B. Folding encurance in weaker direction, Minimum.

CLASS	FOLDS
1	80
2	80
3	80
4	80
5	25
6	200
7	200
8	200
9	200

C. Tearing resistance in weaker direction, Minimum.

CLASS	GRAMS
1	40
2	50
3	60
4	60
5	80
6	30
7	30
8	30
9	30

D. Permanent curl, Maximum, front or back.

CLASS	MILLIMETERS
1	15
2	15
3	15
4	15
5	15
6	25
7	25
8	25
9	25

4. Accelerated Aging Characteristics.

	Maximum Density Change	Minimum Density Change
Light aged samples	25 percent maximum	0.10 maximum
Heat aged samples	20 percent maximum	0.10 maximum

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TABLE II (Continued)

5. Actinic Transmission.

CLASS	BEFORE AGING	AFTER AGING
6	20 percent minimum	17 percent minimum
7	35 percent minimum	25 percent minimum
8	35 percent minimum	25 percent minimum
9	35 percent minimum	25 percent minimum

6. Actinic Opacity.

CLASS	BEFORE AGING	AFTER AGING
6	75 percent minimum	70 percent minimum
7	75 percent minimum	70 percent minimum
8	75 percent minimum	70 percent minimum
9	75 percent minimum	70 percent minimum

3.5.8.4 Developing rate for type 4 papers. The developing rate for type 4 papers shall be such that adequate development of the coating is attained in one pass through the developing section of the apparatus as specified in 4.5.4. Paper of this type shall be developed in a solution certified by the manufacturer for use with type 4 material.

3.5.9 Erasability. Using a slightly gritty eraser, a solid density area 1/4 X 1 inch shall be removed by less than 25 strokes with normal hand pressure. The resulting area shall be free from ghosting and have a smooth surface.

3.6 Uniformity of characteristics. Uniformity of physical and sensitometric characteristics among specimens of the same manufacture and coating number and among different manufacturing lots of a given type, class and speed shall be maintained within closest possible limits.

3.7 Workmanship. Workmanship shall be of the highest grade throughout and shall be in accordance with the best commercial practice in the manufacture of diazo sensitized material.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type, subtype item, and class submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

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4.4.2 Inspection of the end item.

4.2.2.1 Examination of the end item. Examination of the end item shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQL's) specified in 4.2.2.1.5. The lot size for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of rolls or packages of sheets for examination under 4.2.2.1.1, 4.2.2.1.2 and 4.2.2.1.3 and in units of shipping containers for examination if 4.2.2.1.4. Examinations included in 4.4.2.1.1, 4.2.2.1.2 and 4.2.2.1.3 may be performed at any point immediately prior to, during, or after the packaging or spooling operation.

4.2.2.1.1 Examination of the end item for defects in appearance and workmanship. The sample unit for this examination shall be one sheet of sensitized paper or one linear foot of sensitized roll paper as applicable. Not more than one sample unit shall be drawn from any one package or roll.

Examine	Defect
Appearance and Workmanship	Not white paper, cut sheets not square as specified. Paper not completely or evenly coated. Paper contains scuffs, scratches, foreign matter, or abrasions on emulsion surface. Abnormal longitudinal curl. Holes, tears, cuts, cracks or sharp creases.

4.2.2.1.2 Examination of the end item for dimensional defects. The sample unit for this examination shall be one sheet of paper or one roll of paper as applicable. Not more than one sample unit sheet shall be drawn from one package of sheets.

Examine	Defect
Dimensions:	
Sheet Paper:	Length or width varies by more than ± 1/16-inch from specified size.
8 X 10-1/2 through 15 X 18 inches	Length or width varies by more than ± 3/32-inch from specified size.
Sizes larger than 15 X 18 inches	Cut sheets not cut square (see 3.3.1.1).
Roll paper width	Varies by more than ± 1/16-inch from specified width.

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4.2.2.1.3 Examination of the end item for defects in roll construction and length, blocking and count of sheets. The sample unit for this examination shall be one roll or one package of cut sheets, as applicable.

Examine	Defect
Roll Paper	Rolls not evenly or neatly wound; dimensions of core not as specified. Core cracked, broken, chipped, mutilated or collapsed. Not spooled with emulsion side inward (as applicable). Roll not continuous. Roll contains more than permitted number of splices (see 3.5.1.2). Flag at splice missing or not as specified.
Unwinding of roll	Does not unroll evenly and uniformly. Paper blocks during unwinding.
Length of roll	Less than nominal length (see 3.5.1.2 for adjustment for splices).
Count per package of sheets	Number of sheets per package less than specified.

4.2.1.4 Examination of preparation for delivery. An examination will be made to determine that marking, materials, workmanship, contents and weights comply with the requirements of section 5 of this specification. Defects will be scored in accordance with the classification below. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. The lot size shall be the number of shipping containers fully prpeared for delivery, submitted for acceptance at one time.

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Examine	Defect
Packaging and preservation (as applicable)	Not as specified, not in accordance with requirements.
	Flat sheets or rolls (as applicable) not unit wrapped and packaged as specified.
Packing	Unit packages not uniform.
	Containers of improper construction.
	Not packed as specified.
	Packaging material not as specified; closures not accomplished by specified or required methods of materials.
	Number of rolls, boxes or envelopes per carton less than indicated or specified.
Markings (interior) and (exterior)	Interior markings not correct color.
	Data not in accordance with contract specifications.
	Labels (when used) not properly attached.
	Labels or markings illegible, incorrect, incomplete, omitted, or not in accordance with contract requirements.
Weight (Interior and Exterior)	Gross weight exceeds contract requirements.

4.2.2.1.5 Inspection levels and acceptable quality levels (AQL's) for examinations. The inspection levels, for determining the sample size, and the acceptable quality levels (AQL's) expressed in defects per 100 units, shall be as follows:

Examination Paragraph	Inspection Level	AQL
4.2.2.1.1	I	1.0
4.2.2.1.2	S-3	2.5
4.2.2.1.3	S-2	2.5
4.2.2.1.4	S-4	4.0

4.2.3 Testing of the end item. Testing of the end item shall be conducted in accordance with table III for the characteristics as indicated therein on each lot submitted for acceptance. The sample unit, as referenced herein is defined as the amount of material required to perform all tests one time. The sample unit for all testing shall consist of one package of sheet paper or one roll, as applicable. Three sample units, each drawn from a different shipping container, shall be tested with no evidence of failure to meet sample unit or average requirements, as applicable. As an alternate, samples for testing for stock of paper may be drawn at any point prior to the emulsion coating operation. Samples for testing for the sensitometric characteristics, other than fog density may be drawn at any point after the emulsion coating operation. Samples for testing for fog density and the remaining tests shall be drawn at any point not sooner than immediately prior to the beginning of the packaging operation.

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4.3 Apparatus for testing sensitized coating.

4.3.1 Test original. A test original shall be used when indicated in the performance tests. The test original shall be a silver gelatin photographic film. The image area shall have approximately equal transmission at all wave lengths ranging from3500 Angstrom units through 6500 Angstrom units. The test original shall contain one uniform area of silver which has a diffuse transmission density of 0.30 ± 0.05 above fog desnity. Such an area is the second visible step above the clear film of a Kodak No. 3 step tablet.

4.4 Exposure apparatus.

4.4.1 Types 1, 2, 3, and 4 materials. All printing exposures shall be made on a commercial diazo printer calibrated for actual machine speed and having an air-cooled, high pressure mercury vapor, quartz mercury-arc, or high intensity ultraviolet fluorescent light source. The machine shall be equipped with a voltage stabilizer to control voltage fluctuation to the light within ± 5 percent.

4.5 Developing apparatus.

4.5.1 Type 1 material. All tests requiring development, with or without an exposure step, shall be made on the developing section of a commercial ammonia-process printer. Such developing section shall be synchronized in speed with the printer section.

4.5.2 Type 2 material. All tests requiring development, with or without an exposure step, shall be made on the developing section of a commercial moist-process combined printer and developed in which the developed is applied to both sides. Such developing section shall be synchronized in speed with the printer section.

4.5.3 Type 3 material. All tests requiring development, with or without an exposure step, shall be made on the developing section of a commercial moist-process combined printer and developed in which the developer is applied to both sides. Such developing section shall be synchronized in speed with the printer section.

4.5.4 Type 4 material. All tests requiring development, with or without an exposure step, shall be made on the developing section of a machine certified by the manufacturer for use with type 4 material.

4.6 Developing.

4.6.1 Type 1 material. All developing for those tests requiring development, but not exposure, shall be accomplished on a commercial ammonia-process machine in good operating condition, at a speed corresponding to relative printing speed for the material as determined in 4.13 and setting for ammonia and temperature recommended by the manufacturer. Development for those tests requiring exposure prior to development shall be accomplished at the same speed as that used for exposure on machines having synchronized printer-developer speeds.

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TABLE III. Instructions for testing (sample unit)

Characteristic	Specification Reference		Requirements Applicable to		Number Determinations per sample unit	Results Reported as	
	Requirement	Test Method	Indiv. Unit	Lot Average		Pass or Fail	Numerically to nearest
Physical Characteristics:							
*Basis weight.....	3.5.2	4.9.1	.....	X	1 (Avg of 10 shts)	.....	0.1 pound
*Folding endurance in weaker direction	3.5.2	4.9.1	.....	X	Avg of 10	.....	number of double folds
*Tearing resistance in weaker direction	3.5.2	4.9.1	.....	X	Avg. of 5	.....	grams
*Stock	3.4.4	4.9.1	.....	.....	Avg. of 2 on composite	X	5 percent
Temporary Curl	3.5.2.1	4.10	X	.....	1	X	.....
Permanent Curl (face or back)	3.5.2.2	4.11	X	.....	Avg. of 2	.....	millimeter
Package curl	3.5.2.3	4.10	X	.....	1	X	.....
Coating defects after exposure and processing	3.5	4.12	X	.....	1	X	.....
Speed rating	3.4.6	4.13	X	.....	Avg of 2	.....	1.0 speed
Developing rate: Type 1	3.5.8.1	4.5.1	X	.....	1	X	.....
Developing rate: Type 2	3.5.8.2	4.5.2	X	.....	1	X	.....
Developing rate: Type 3	3.5.8.3	4.5.3	X	.....	1	X	.....
Developing rate: Type 4	3.5.8.4	4.5.4	X	.....	1	X	.....
Maximum image density	3.4.7	4.14.1	X	.....	Avg. of 2	.....	.01 density
Minimum background density	3.4.8	4.14.2	X	.....	Avg of 2	.....	.01 density
Maximum image density of faded samples	3.5.3.1	4.14.3	X	.....	Avg. of 2	.....	.01 density
Minimum background density of faded samples	3.5.3.2	4.14.4	X	.....	Avg of 2	.....	.01 density
Maximum image density of aged samples	3.5.3.3	4.14.5.1	X	.....	Avg of 2	.....	.01 density
Minimum background density of aged samples	3.5.3.4	4.14.5.2	X	.....	Avg of 2	.....	.01 density
Actinic opacity aged and unaged as applicable	3.5.5	4.15	X	.....	Avg of 2	.....	% opacity
Actinic transmission, as as applicable, aged and unaged.	3.5.4	4.16	X	.....	Avg of 2	.....	% transmission

*These tests shall be performed on first production lot and may be certified for succeeding lots.

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4.6.2 Types 2 and 3 materials. All developing for those tests requiring development, but not exposure, shall be accomplished on a commercial moist-process machine, in good operating condition, in which the developer is applied to both sides at a speed corresponding to the relative printing speed for the material as determined in 4.13. Development for those tests requiring exposure prior to development shall be accomplished at the same speed as that used for exposure on machines having synchronized printer-developer speeds. All prints of type 2 and 3 materials shall be developed in the solution recommended by the manufacturer.

4.6.3 Type 4 material. All developing for those tests requiring development, but not exposure, shall be accomplished on a machine certified by the manufacturer for use with type 4 material, in good operating condition. The developer is to be applied at a speed corresponding to the relative printing speed for the material as determined in 4.13. Development for those tests requiring exposure prior to development shall be accomplished at the same speed as that used for exposure on machines having synchronized printer-developer speeds. All prints of type 4 materials shall be developed in the solution recommended by the manufacturer.

4.7 Reflection densitometer. A densitometer conforming to ANSI Standard PH2.17 shall be used. The densitometer shall be equipped with S-4 response phototube, a Wratten 106 filter and an IR 4 filter and it shall be adjusted to read "0" for 100 percent reflectance. This may be accomplished by using a fresh magnesium oxide surface (see NBS Circular 547) which, if properly prepared, will have a reflectance of 100 percent for such a densitometer. A more convenient standard which has been calibrated against magnesium oxide with a reflectance between 78 percent and 100 percent (density 0.00 to 0.108) may be used for setting the low end of the scale. A calibrated standard of approximately 1 percent reflectance (density 2.00) shall be used for checking the high density end of the scale.

4.8 Fade-O-Meter. The Fade-O-Meter shall be Model 18 or equivalent as manufactured by the Atlas Electrical Devices Company. All tests made with this apparatus shall be conducted with a black panel temperature of 90°F. and the humidifier switch "ON".

4.9 Test methods.

4.9.1 Physical tests. The following tests shall be made in accordance with the listed TAPPI methods.

Test	Method No.
Atmospheric conditions	T402
Basis weight	T410
Folding endurance	T511
Tearing resistance	T414
Fiber identification and quantitative determination	T401

4.10 Temporary curl. Twenty-five sheets 8-1/2 X 11 inches of each material to be tested shall be run, under a paper master, continuously at the correct speed with the grain direction parallel to that of travel. The paper master shall have a minimum actinic transmission of 35 percent. Samples stacking satisfactorily shall be deemed satisfactory.

4.11 Permanent curl. Select at random two sheets of the fully processed prints of the material under test. Place the two sheets on a flat surface, one print side up and the other print side down, in a conditioning chamber capable of maintaining a relative humidity of 50 percent ± .05 percent and a temperature of 70° ± 30°F. Allow the prints to remain in the chamber at approximately 70°F. and 50 percent relative humidity for a period of 24 hours. At the end of this period, measure and record in millimeters the greatest lift from the flat surface of any part of either sheet. Report the curl as millimeters face or back curl, as applicable.

4.12 Coating defects. One sample of each type and subtype of paper specified herein, not less than an 8-1/2 by 11 cut sheet nor less than one linear yard in length and the full width or roll material, shall be selected at random from the sensitized paper to be tested. Each of the samples shall be cut

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into thirds. Two thirds of all the samples shall be put aside protected from exposure to light. The unprotected third shall be examined in visible-light to determine conformance with requirements specified in 3.5.7. One third of the protected samples shall be flash exposed and fully processed on applicable equipment for each subtype of sensitized paper. These samples (after exposure and processing) shall be held not more than six feet from an unshielded 60 watt incandescent lamp and again visually examined for compliance with requirements specified in 3.5.7. The final protected third of all samples shall be exposed, using a test film containing parallel straight lines and spaces of equal width, and processed on applicable equipment. These samples (after exposure and processing) shall be visually examined in white light for coating streaks or other defects which would impair print quality.

4.13 Speed rating.

4.13.1 ARMM sensitometer. The speed ratings of diazo sensitized papers specified herein shall be determined using an ARMM sensitometer or equivalent and a densitometer as specified in 4.7 as follows:

After the requisite warm-up and calibration of the sensitometer, exposures are made on the test papers for various lengths of time until a density of 0.20 to 0.40 is obtained. Care must be taken to insure complete development on the equipment recommended by the paper manufacturer. Using the exposure time that produced the 0.20 to 0.40 density as a starting point, expose a series of test papers increasing each exposure time by a factor of 10 percent over the previous exposure time until burnout density is reached. (Burnout density is the minimum density of the product with complete exposure.) Measure the density of the exposed and developed samples. Plot the points on coordinate paper using the times of exposure in seconds for the abscissa (x axis) and the corresponding densities as the ordinate (y axis) and draw a smooth curve of best fit through the points. From the curve, accurately determine the exposure time in seconds to expose the sample to a density of 0.05 above burnout density. From this time and the sensitometer constants, the energy to expose the sample to a density of 0.05 above burnout is calculated in units of watt-seconds per square centimeter. The reciprocal of this value is calculated and the ARMM speed rating is found on Graph I.

4.13.2 Alternate ARMM speed rating method. A diazotype paper standard which has been assigned an ARMM speed rating shall be used. A package of the speed standard shall be removed from cold storage and allowed to come to room temperature before opening. The speed standard shall be exposed in contact with the test original at a machine speed resulting in print density, under the image areas of the test original, of 0.05 $\pm$ 0.01 above burnout. The machine speed in feet per minute at which these conditions are met shall be the print speed of the standard. Determine the print speed of the sample by taking a sheet from a previously unopened package and immediately exposing it in a similar manner. Calculate its speed rating as follows:

$$\frac{\text{Print speed of sample}}{\text{Print speed of standard}} \times \text{ARMM Standard speed rating} = \text{Sample speed rating.}$$

4.14 Density.

4.14.1 Maximum density. Completely develop, without exposure, two sheets of material under test, not less than 8-1/2 X 11 inches each, on a commercial diazotype processor in good working condition. For type 1 material, air the developed sheets for at least fifteen minutes to allow excessive ammonia vapors to escape. Using a visual filter (Wratten 106 or equivalent) check the zero and sensitivity settings of the densitometer with the appropriate working standards. Take five density readings across the diagonal on the image side at two inch intervals. For this test use an opaque white backing sheet with a brightness of 84 percent $\pm$ .01 percent. Determine and report the average of the readings as maximum reflection density.

4.14.2 Minimum density. Completely expose two sensitized sheets so that all the sensitized material will be entirely bleached or decomposed. The sheets shall be developed and density measurement shall be taken at two-inch intervals on a diagonal of each of the sheets. The average of not less than six measurements so taken shall be minimum density.

4.14.3 Maximum density of faded samples. Two section shall be cut from the samples used in the maximum image density test (see 4.14.1) and exposed in an Atlas Fade-O-Meter, Type FDE-R or equal, for 3 hours at 90°F. black panel temperature with humidifier switch "ON". After this exposure the samples shall be conditioned to 70°F. and 50 percent relative humidity and the maximum density shall be redetermined.

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4.14.4 Minimum density of faded samples. Two sections shall be cut from the samples used in the minimum density test (see 4.14.2) and exposed in the Fade-O-Meter for 3 hours at 90°F. black panel temperature with humidifier switch "ON". After this exposure the samples shall be conditioned to 70°F. and 50 percent relative humidity, and the minimum density shall be redetermined.

4.14.5 Aging characteristics. An unopened production package of the smallest quantity normally made with an overwrap of aluminum foil, taped completely to seal the package from air is placed in an oven which has reached a temperature of 50° ± 1°C. The package is kept in the over at 50° ± 1°C for a period of five days. At the end of the test period the package is removed and allowed to cool to room temperature.

4.14.5.1 Maximum density of aged samples. The maximum density of the aged samples shall be determined by procedure specified in 4.14.1.

4.14.5.2 Minimum density of aged samples. The minimum density of the aged samples shall be determined by procedure specified in 4.14.2.

4.15 Actinic opacity. The actinic opacity of the maximum density area of a diazo type reproducible is determined as outlined below:

1. Develop completely without exposure a sheet of slow-speed reprint paper 8-1/2 X 11 inches in size. This produces the maximum achievable density of the reprint paper. Use a reprint paper of Image Color 1 and AEEM speed 91-115 for type 1 reproducibles and a reprint paper of Image Color 2 and AEEM speed 51-70 for types 2 and 3 reproducibles.

2. Cover one half of a sheet of the reproducible under test with a completely opaque material such as aluminum foil. Then expose and develop the reproducible at a print speed that produces a faintly discernible haze in the uncovered half of the sheet.

3. With the print obtained in step 2 as the master, expose and develop another sheet of the reprint paper at a speed that produces a background density of 0.02 ± 0.01 above burnout density.

4. Using the instrument specified in paragraph 4.7, the density value of the reprint obtained in step 1 is measured and designated as "Maximum Density of Reprint Paper." Then the density value of the reprint obtained in step 3 is measured and designated as "Maximum Reprint Density."

5. The actinic opacity of the reproducible equals:

$$\text{Actinic Opacity} = \frac{\text{Maximum reprint Density}}{\text{Maximum Density of Reprint Paper}} \times 100$$

4.16 Actinic transmission. The actinic transmission of nonimage (burn-out density) area of a diazotype reproducible is determined as outlined below.

1. Without the use of a master, expose and develop a sheet or AEEM speed 51-70 reprint paper at such a speed as to leave a clearly discernible amount of haze in the background. Record the true linear machine print speed as "Maximum Print Speed of Reprint Paper."

2. Without the use of a master, expose and develop a sheet of the reproducible under test at such a speed as to completely burnout all the light sensitive diazo coating, leaving a clean background.

3. Using this fully exposed and developed reproducible as the master, expose and develop a sheet of reprint paper at such a speed as to leave a background haze equal to that obtained in step 1. Record the true linear machine speed as "Reprint Speed of the Recproducible."

4. The percent actinic transmission of the reproducible equals.

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$$\text{Actinic} = \frac{\text{Reprint speed of reproducible (Step 3)}}{\text{Transmission Maximum print of reprint paper (Step 1)}} \times 100$$

5. PREPARATION FOR DELIVERY

5.1 Packaging. (Packaging shall be level A or C, as specified (see 6.1.2).

5.1.1 Level A. (Sheets and rolls). Sheets of sensitized paper shall be packaged in units as specified (see 6.1.2). Unless otherwise specified, the sheets or rolls shall be securely wrapped in such a manner as to protect the diazo sensitized papers from moisture, light and handling. The package of sheets or rolls shall be sealed with tape conforming to PPP-T-60 or in such a manner as to protect the diazo sensitized papers from moisture, light and handling. Each package or roll of diazo sensitized paper shall be preserved in accordance with method IC of MIL-P-116.

5.1.2 Level C. Sheets and rolls of sensitized paper shall be packaged in accordance with the manufacturer's standard practice, providing that this will insure protection for the paper during shipment and provides for safe delivery to its destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1.2).

5.2.1 Level A. The paper packaged as specified in 5.1,shall be packed as specified for level A in MIL-P-3447.

5.2.2 Level B. The paper packaged, as specified in 5.1,shall be packed as specified for level B in MIL-P-3447.

5.2.3 Level C. The paper packaged, as specified in 5.1, shall be packed in containers that will assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification rules and the National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Unit package marking. Unless otherwise specified (see 6.1.2) commercial labeling will be acceptable providing that the label includes in ledgible characters the following information:

Manufacturer's name.

Trade name of product.

Classification designation.

Federal Stock Number.

Size and number of sheets or width and length of rolls, subtype, class, speed coating number and the expiration date.

5.3.2 Intermediate and shipping container marking. In addition to the special marking required by 5.3.3 all marking shall be in accordance with 5.3.2.1, as specified (see 6.1.2).

5.3.2.1 Civil agencies and military activities. In addition to any special marking required by the contract or order, all intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129 for military activities and Fed. Std. No. 123 for Civil activities.

5.3.3 Special marking. In addition to the markings required by 5.3.1 or 5.3.2, the following special marking shall be on all containers:

STORE IN COOL DRY LOCATION AWAY FROM AMMONIA VAPORS AND SUNLIGHT

This marking shall be located where it will not be obliterated when the packages are opened.

6. NOTES

6.1 Intended use. The diazotype paper covered by this specification is intended for use in making reproducibles, and reproductions of original drawings, tracings and other documents.

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6.1.1 Supersession data. This specification supersedes the following Federal and Military specifications. Requirements of the listed superseded documents are included in this specification.

Federal Specification UU-P-221A, October 27, 1964

MIL-P-9853 (USAF), May 19, 1959

MIL-P-4201 (USAF), January 12, 1951

6.1.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Stock number.
(c) Classification designators (see table I).
(d) Roll, sheet, widths and lengths (see 3.5.1.1 and 3.5.1.3).
(e) Sizes and number of sheets per box (see 5.1).
(f) Selection of samples (see 4.2).
(g) Type of packaging and packing required (see 5.2).
(h) Preproduction samples and quantity required (see 3.3.1).
(i) Special markings (see 5.3).

6.2 Definitions.

6.2.1 ARMM. Association of Reproduction Materials Manufacturers.

6.2.2 ARMM speed standard. A diazotype product calibrated with the ARMM approved sensitometer and designed to serve as a reference standard for speed rating determination. The speed standard is available from the Association of Reproduction Materials Manufacturers, 901 North Washington Street, Alexandria, Virginia 22314.

6.2.3 ARMM sensitometer. A sensitometer designed and developed for the reproduction industry by the Association of Reproduction Materials Manufacturers, for the precise and reproducible measurements of the light sensitivity of diazotype materials.

6.2.4 Print speed. The exposure in feet per minute in the apparatus specified in 4.4.1, as determined in 4.13. This value is a relative speed rating. For example, a product with a speed rating of 100 will have a print speed twice the speed of a product with a speed rating of 50.

6.2.5 Translucent paper. A paper reproducible which is made in its original form in such a manner as to transmit a relatively high percentage of light.

6.2.6 Transparentized paper. A translucent paper impregnated to render it capable of transmitting a higher percentage of light.

6.3 Transportation description. Transportation description and minimum weights applicable to this commodity are:

RAIL:

Paper, blueprint or reproduction not otherwise indexed by name, sensitized.

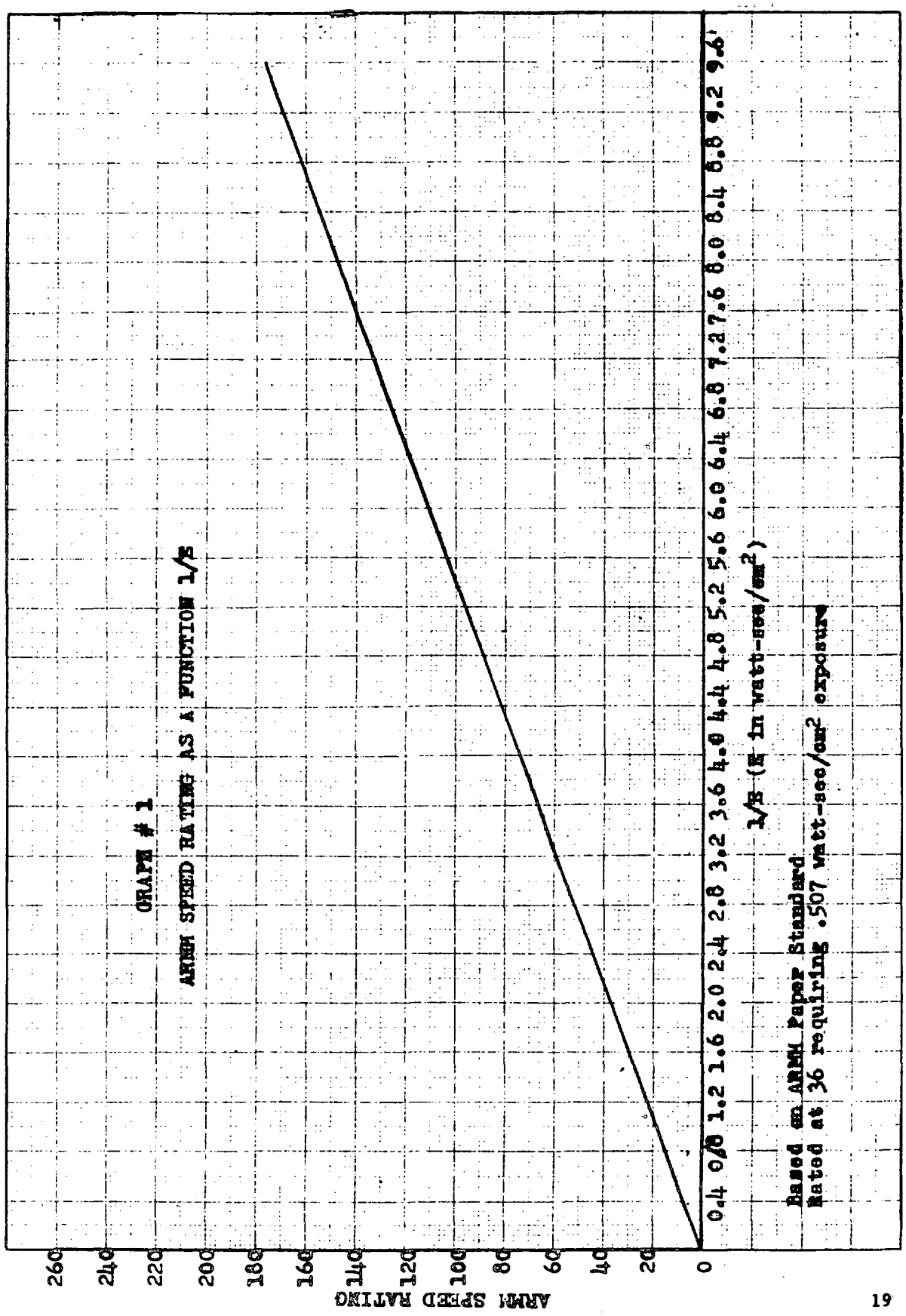
Carload minimum weight 36,000 pounds.

MOTOR:

Paper, blueprint or reproduction not otherwise indexed sensitized.

Truckload minimum weight 36,000 pounds subject to Rule 115 National Motor Freight Classification.

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Army - EL, MI, WC

CIVIL AGENCY INTEREST:

GSA - FSS

Preparing activity:

GSA-FSS

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